COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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ACTIVITY NO. _8.1_ Using Sorting Algorithms 2

1. PROCEDURE OUTPUT

1. Explain the quick sort, shell sort, and merge sort types of sorting algorithms.
    - **Quick Sort**
        - Quick sort selects a pivot and uses two pointers, i and j, to check and swap misplaced elements, moving larger values right and smaller ones left. Once the scan completes, a final swap places the pivot into its exact sorted position at index j, ensuring all elements to its left are smaller and to its right bigger.
    - **Shell Sort**
        - Shell sort breaks the O(n^2) time complexity barrier for sorting. Shell sort functions first by comparing the elements that have a wide gap then it will gradually reduce the gap. This helps by allowing the element of the array to move faster. In short, it chooses a gap sequence, sorts the elements at each gap and reduces the gap and it repeats until the gap becomes 1.
    - **Merge Sort**
        - In merge sort, we have 2 functions. First is the mergeSort which divides the array into two until it only has one element. Then the merge function merges the two sub arrays while comparing the elements of which should come first. Working backwards until we get the full array sorted. This sorting algorithm also begins by sorting the left side of the array before working on the other side. Finally, this sorting algorithm has a time complexity of O(n log n)

2. Give simple sample programs in C++ that use the above sorting algorithms. Use a user input of 10 integer values in your example elements in an array to be sorted. Explain how the programs work.
    - **Quick Sort**
        - A C++ program implementing quick sort can be seen in gaming leaderboard applications that rank player scores. The program accepts 10 user-entered scores into an array and uses two pointers (i and j) to partition the values around a pivot element, swapping smaller and larger scores to opposite sides before recursively sorting each section in place to output the final 10 rankings.
    - **Shell Sort**
        - A sample program that uses shell sort is the uClibc library which is a standard C library that is designed for embedded Linux systems such as routers and microcontrollers. Since ram and stack memory are very small, the uClibc library implements the shell sort to safely sort the arrays directly on the chip rather than using the quick sort, risking the device's call stack.
    - **Merge Sort**
        - A music streaming app would benefit from a merge sort since it has multiple parameters.

The order of the sorted list would mostly stay the same in the case that parameters were changed. Let's say we sorted the list by the year of publishing, then sorted it again by artists, since all the songs are already in chronological order, songs of one artist would also already be in chronological order since they come more to the left of the array than the other songs.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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